PIND 167326

MEETING MINUTES

Eric Vallabh Minikel, PhD
Group Leader, Broad Institute
[redacted]
Cambridge, MA 02142

Dear Dr. Minikel:[1]

Please refer to your pre-investigational new drug application (PIND) file for PrP-lowering divalent siRNA 2439-exNA.

We also refer to the video conference between representatives of your firm and the FDA on June 30, 2023. The purpose of the meeting was to discuss a path forward for a small investigator-initiated trial of 2439-exNA in symptomatic patients diagnosed with prion disease.

A copy of the official minutes of the video conference is enclosed for your information. Please notify us of any significant differences in understanding regarding the meeting outcomes.

If you have any questions, please contact Roshell Weatherless-Stroble, Regulatory Project Manager at [redacted] or at [redacted].

Sincerely,

{See appended electronic signature page}

Emily Freilich, MD
Director
Division of Neurology 1
Office of Neuroscience
Center for Drug Evaluation and Research

Enclosure:
- Meeting Minutes

[1] We update guidances periodically. For the most recent version of a guidance, check the FDA Guidance Documents Database https://www.fda.gov/RegulatoryInformation/Guidances/default.htm.

U.S. FOOD & DRUG ADMINISTRATION

MEMORANDUM OF MEETING MINUTES

Meeting Type: B
Meeting Category: Pre-IND

Meeting Date and Time: June 30, 2023, from 11:00 AM to 12:00 PM ET
Meeting Location: Video conference

Application Number: PIND 167326
Product Name: PrP-lowering divalent siRNA 2439-exNA
Indication: For the treatment of symptomatic patients with prion disease

Sponsor Name: Eric Vallabh Minikel, PhD
Regulatory Pathway: 351(a) of the Public Health Service Act

FDA ATTENDEES

Office of Neuroscience
Teresa Buracchio, MD, Director (Acting)

Division of Neurology 1
Emily Freilich, MD, Director
Michael Dimyan, MD, Clinical Reviewer

Division of Pharmacology and Toxicology for Neuroscience
Lois Freed, PhD, Director
Chun-Ting Lee, PhD, Nonclinical Reviewer

Office of Pharmaceutical Quality (OPQ)
Martha Heimann, PhD, Senior Pharmaceutical Quality Assessor
Mariappan Chelliah, PhD, Drug Product Reviewer

Office of Clinical Pharmacology
Bilal AbuAsal, PhD, Team Leader
Min Li, PhD, Reviewer
Hobart Rogers, PharmD, PhD, Genomics and Targeted Therapy Reviewer

Division of Regulatory Operations for Neuroscience
Roshell Weatherless-Stroble, PharmD, Regulatory Project Manager

SPONSOR ATTENDEES
Sonia Vallabh, PhD, Group Leader
Eric Minikel, PhD, Group Leader

Eric Lander, PhD, Founding Director Emeritus
Steven E. Arnold, MD, Professor of Neurology
Brian S. Appleby, MD Director, National Prion Disease Pathology Surveillance Center
Alissa Coffey, Project Manager, Broad Institute
Lauren Black, Nonclinical Consultant, Charles River Laboratories
Kevin Fettes, CMC Consultant, FTS Pharma Consulting LLC

1.0 BACKGROUND

Dr. Eric Vallabh Minikel of the Broad Institute of MIT and Harvard submitted a meeting request to discuss a path forward for a small investigator-initiated trial of prion protein (PrP) lowering divalent siRNA 2439-exNA in symptomatic patients diagnosed with prion disease.

The sponsor states that the therapeutic hypothesis of PrP-lowering is supported by decades of genetic experiments in animal models and more recently by preclinical proof-of-concept studies in mice using antisense oligonucleotides, but to date, no therapy exists, and no PrP-lowering drug candidate has ever entered human trials.

Broad Institute has had previous interactions with the Agency including a Critical Path Innovation Meeting (CPIM) to discuss their preliminary work to develop a therapeutic strategy for genetic prion disease (November 14, 2017).

FDA sent preliminary comments to Dr. Eric Vallabh Minikel on June 28, 2023.

2.0 DISCUSSION

2.1. Clinical

Question 1: We propose to test a single dose in symptomatic patients, escalating dose after an independent committee evaluates safety data, but not waiting for biofluid analysis results to become available. Is the proposed dose escalation schedule and monitoring program adequate?

FDA Response to Question 1:
It is premature to comment on the proposed dose escalation plan as your product development of 2439-exNA, including nonclinical toxicology studies, is incomplete. Please see response to Question 2.

In general, the selection of a safe starting dose in humans, the dose escalation schedule, and safety monitoring should be based on results of the nonclinical studies and will be a matter of review at the time of IND submission. Your IND submission should include a comprehensive discussion regarding potential adverse effects of PRP depletion and propose an appropriate monitoring strategy.

You will need to conduct an in vitro drug interaction study to characterize the DDI risk related to divalent siRNA 2439-exNA as a perpetrator for major metabolic enzymes and transporters before you initiate the first-in-human study in prion disease patients.

Discussion: None.

2.2. Preclinical

Question 2: We propose to conduct GLP toxicology and pharmacokinetic studies in rats, supplemented by non-GLP pharmacokinetic/pharmacodynamic studies in mice.

Does the agency agree that data from these studies would be appropriate to support the proposed clinical plan?

FDA Response to Question 2:
Nonclinical safety testing is typically conducted in two species (rodent and nonrodent). In addition, we have concerns regarding the potential toxicity of divalent siRNA 2439-exNA, which has modifications (e.g., exNA) that may potentially increase its toxicity and for which you state there is no previous human experience. Therefore, to support the proposed clinical study of siRNA 2439-exNA, you will need to provide GLP-compliant toxicity studies of divalent siRNA 2439-exNA in rodent and nonrodent.

The pivotal toxicity studies in rodent and nonrodent should be conducted in an adequate number of animals, i.e., at least 10/sex/group and 4/sex/group, respectively, for the main study, with an additional 5/sex/group and 2/sex/group, respectively, if recovery is to be assessed. The studies should be of duration similar to or greater than (e.g., up to one month) that proposed for humans and test three dose levels of siRNA 2439-exNA in addition to controls, using the clinical route and a clinically relevant dosing regimen. Justification should be provided in each study report for dose and species selection. We refer you to guidance for a general discussion of the criteria for dose selection (ICH M3(R2), January 2010, ICH M3(R2) Q&A, February 2013).

The core safety pharmacology endpoints (central nervous system, cardiovascular, and respiratory) should be assessed in separate, dedicated studies or, preferably, incorporated into the appropriate GLP-compliant toxicity studies.

The adequacy of the nonclinical studies will be a matter of review.

Discussion:

To support a single-dose study in humans, the sponsor proposes to conduct GLP-compliant single-dose intrathecal (IT) toxicity studies in rat and nonrodent (dog or sheep). The Division stated that single-dose toxicity studies would be sufficient to

support a single-dose clinical study and agreed with the general design of the studies, based on the limited information provided, but had the following additional comments:

- For both rodent and nonrodent, animal should be sacrificed sooner (e.g., 24 hours) after dosing than specified.
- Assessment of recovery is optional but may be important, depending on the toxicity(ies) observed in main-study animals. Additional animals (5/sex/group and 2/sex/group for rodent and nonrodent, respectively) should be included if recovery is to be assessed. A 1-month recovery period would be reasonable.
- Justification should be provided for the species selected for the pivotal studies, taking into consideration the availability of data to document their relevance to humans for assessing hybridization and nonhybridization-mediated toxicity. For example, there may be less experience with nonrodent species such as dog or sheep compared to nonhuman primate or minipig.

Although single-dose toxicity studies would be sufficient, the Division suggested the sponsor consider longer duration toxicity studies in order to allow greater flexibility in the administration of siRNA 2439-exNA to humans.

2.3. Chemistry Manufacturing and Controls (CMC)

Question 3: We propose to generate Drug Product vials of different sizes, with smaller aliquots used for testing than dosing, to maximize utilization of manufactured drug.

Does the agency agree that this plan is acceptable?

FDA Response to Question 3:

Your proposal may be acceptable for the initial study if the vials used for stability testing (and other quality control tests) can represent the worst case in terms of the head space - i.e., compared with the vials used for clinical dosing, such that the stability vials have higher headspace to drug product fill volume ratio. Final determination of the acceptability of such approach would be made at the time of the IND submission when all the supportive data may be evaluated.

Please note that as the drug development continues, you should use the same container closure system for all the drug product vials.

Discussion:

- On slide 19 of the PowerPoint presentation, the sponsor proposed to use the same vial for clinical use and quality control testing, but the latter will have reduced fill volume.

    - The Agency responded that the proposal appeared reasonable as the reduced fill volume of the quality control vials will have higher headspace, and therefore will represent the worst-case scenario.
- The sponsor's proposal of sterile fill-and-finish were discussed on slide 22 and 23 of the PowerPoint presentation.
    - The sponsor proposed exemption on the topic of qualification/verification/validation (BNF) of USP 71, 61, 85 methods for sterility, endotoxin, bioburden.
        - As there was no micro data available at the time of this meeting package, the Agency could not comment on the micro questions and recommended the sponsor submit micro questions in a separate meeting package for feedback. Otherwise, the micro data will be assessed at the time of the IND safety review.
    - Particulates
        - The Agency commented that the proposal of using 1 vial for the particulate testing may be acceptable at the beginning of the IND studies. However, as the drug development proceeds, more vials would need to be tested as per USP 788.
    - In-process Bioburden
        - The Agency could not comment on the micro questions and recommended the sponsor submit micro questions in a separate meeting package for feedback. Otherwise, the micro data will be assessed at the time of the IND safety review.
    - Stability
        - In response to the Sponsor's proposal to conduct stability testing at 6, 12, 24 months at -20°C, the Agency asked to include a 3-month time point in the testing.
        - The Sponsor inquired about the minimum stability data that needs to be included in the IND at the time of filing. The Agency stated that the submission should include at least three months of long-term (-20°C) or 1 month of accelerated stability data (refrigerated storage condition) for the proposed clinical batch. In lieu of stability data for the clinical batch, the Agency may accept the data from a representative batch if it is manufactured using the same process as that of the clinical batch. However, stability of the clinical batches should be assessed in parallel with the clinical study, at least for the duration of the clinical study.
        - In response to the Sponsor's proposal to omit in-use stability testing, the Agency stated that there is no data available on how stable the drug product would be once it is diluted and stored at room temperature. Therefore, the acceptability of the lack of the in-use stability for the IND opening will depend on the totality of the data, including how long the admixture will be stored at room temperature and it will be an IND review issue. Furthermore, if the

admixture will be stored for more than 4 hours at room temperature, microbial attributes also should be assessed.

Additional Discussion:

The sponsor proposed to submit an expanded access IND rather than standard research or commercial IND.

The Agency informed the sponsor that this would not be considered as an expanded access IND and that a research or commercial IND would be more appropriate. The Agency further explained that expanded access is reserved for drug products under an existing IND where there is clinical and nonclinical data that could be cross referenced. The sponsor indicated that this IND would not be commercialized and would therefore be research.

The Agency stated that the N-of-1 guidances are not applicable to this development program as these guidances are intended for the development of an ASO to treat a small number of individuals (usually one or two) who have a unique genetic variant that may be able to benefit from an individualized ASO.

The Agency stated the regulatory pathway for this IND would be discussed internally and provided in a post-meeting comment.

Post-meeting Comment:
In general, treatment use under expanded access is authorized when a drug has undergone some clinical development, has shown some potential for benefit, and there is a request to “expand” access to the drug to patients who are not eligible to participate in ongoing trials, cannot access trial sites, or when clinical trials are no longer ongoing. That is not the case here and, as we previously indicated, we believe a research IND to be the more appropriate approach for your submission. Because there are a limited number of patients with prion disease in the United States, it is important to patients who now have, or in the future will be diagnosed with, prion disease, to determine whether there is substantial evidence of effectiveness of your product for the treatment of the disease. Therefore, we recommend that you develop a plan to assess the safety and efficacy of your product, including the conduct of an adequate and well-controlled study. We are willing to work with you on the design of such a development program.

3.0 ADDITIONAL INFORMATION

PREA REQUIREMENTS

Under the Pediatric Research Equity Act (PREA) (21 U.S.C. 355c), all applications for new active ingredients (which includes new salts and new fixed combinations), new indications, new dosage forms, new dosing regimens, or new routes of administration

are required to contain an assessment of the safety and effectiveness of the product for the claimed indication(s) in pediatric patients unless this requirement is waived, deferred, or inapplicable.

Please be advised that under the Food and Drug Administration Safety and Innovation Act (FDASIA), you must submit an Initial Pediatric Study Plan (iPSP) within 60 days of an End-of-Phase-2 (EOP2) meeting. In the absence of an EOP2 meeting, refer to the draft guidance below. The iPSP must contain an outline of the pediatric study or studies that you plan to conduct (including, to the extent practicable study objectives and design, age groups, relevant endpoints, and statistical approach); any request for a deferral, partial waiver, or waiver, if applicable, along with any supporting documentation, and any previously negotiated pediatric plans with other regulatory authorities. The iPSP should be submitted in PDF and Word format. Failure to include an Agreed iPSP with a marketing application could result in a refuse to file action. For additional guidance on the timing, content, and submission of the iPSP, including an iPSP Template, please refer to the draft guidance for industry *Pediatric Study Plans: Content of and Process for Submitting Initial Pediatric Study Plans and Amended Pediatric Study Plans*.[2] In addition, you may contact the Division of Pediatric and Maternal Health at 301-796-2200 or email [redacted]. For further guidance on pediatric product development, please refer to FDA.gov.[3]

DATA STANDARDS FOR STUDIES

Under section 745A(a) of the FD&C Act, electronic submissions "shall be submitted in such electronic format as specified by [FDA]." FDA has determined that study data contained in electronic submissions (i.e., NDAs, BLAs, ANDAs and INDs) must be in a format that the Agency can process, review, and archive. Currently, the Agency can process, review, and archive electronic submissions of clinical and nonclinical study data that use the standards specified in the Data Standards Catalog.[4]

On December 17, 2014, FDA issued the guidance for industry *Providing Electronic Submissions in Electronic Format - Standardized Study Data.* This guidance describes the submission types, the standardized study data requirements, and when standardized study data are required. Further, it describes the availability of implementation support in the form of a technical specifications document, *Study Data Technical Conformance Guide*, as well as email access to the eData Team ([redacted]) for specific questions related to study data standards. Standardized study data are required in marketing application submissions for clinical and nonclinical studies that started after December 17, 2016. Standardized study data are required in commercial IND application submissions for clinical and nonclinical

[2] When final, this guidance will represent the FDA's current thinking on this topic. For the most recent version of a guidance, check the FDA guidance web page at https://www.fda.gov/RegulatoryInformation/Guidances/default.htm.
[3] https://www.fda.gov/drugs/development-resources/pediatric-and-maternal-health-product-development
[4] http://www.fda.gov/forindustry/datastandards/studydatastandards/default.htm

studies that started after December 17, 2017. CDER has produced a Study Data Standards Resources web page[5] that provides specifications for sponsors regarding implementation and submission of clinical and nonclinical study data in a standardized format. This web page will be updated regularly to reflect CDER's growing experience in order to meet the needs of its reviewers.

For commercial INDs and NDAs, Standard for Exchange of Nonclinical Data (SEND) datasets are required to be submitted along with nonclinical study reports for study types that are modeled in an FDA-supported SEND Implementation Guide version. The FDA Data Standards Catalog, which can be found on the Study Data Standards Resources web page noted above, lists the supported SEND Implementation Guide versions and associated implementation dates.

Although the submission of study data in conformance to the standards listed in the FDA Data Standards Catalog will not be required in studies that started on or before December 17, 2016, CDER strongly encourages IND sponsors to use the FDA supported data standards for the submission of IND applications and marketing applications. The implementation of data standards should occur as early as possible in the product development lifecycle, so that data standards are accounted for in the design, conduct, and analysis of clinical and nonclinical studies. For clinical and nonclinical studies, IND sponsors should include a plan (e.g., in the IND) describing the submission of standardized study data to FDA. This study data standardization plan (see the FDA Study Data Technical Conformance Guide) will assist FDA in identifying potential data standardization issues early in the development program.

If you have not previously submitted an eCTD submission or standardized study data, we encourage you to send us samples for validation following the instructions at FDA.gov. For general toxicology, supporting nonclinical toxicokinetic, and carcinogenicity studies, submit data in the Standards for the Exchange of Nonclinical Data (SEND) format. The validation of sample submissions tests conformance to FDA supported electronic submission and data standards; there is no scientific review of content.

The Agency encourages submission of sample data for review before submission of the marketing application. These datasets will be reviewed only for conformance to standards, structure, and format. They will not be reviewed as a part of an application review. These datasets should represent datasets used for the phase 3 trials. The FDA Study Data Technical Conformance Guide (Section 7.3 eCTD Sample Submission pg. 44) includes the link to the instructions for submitting eCTD and sample data to the Agency. The Agency strongly encourages Sponsors to submit standardized sample data using the standards listed in the Data Standards Catalog referenced on the FDA Study Data Standards Resources web site. When submitting sample data sets, clearly identify them as such with **SAMPLE STANDARDIZED DATASETS** on the cover letter of your submission.

[5] http://www.fda.gov/ForIndustry/DataStandards/StudyDataStandards/default.htm

Additional information can be found at FDA.gov.[6]

LABORATORY TEST UNITS FOR CLINICAL TRIALS

CDER strongly encourages IND sponsors to identify the laboratory test units that will be reported in clinical trials that support applications for investigational new drugs and product registration. Although Système International (SI) units may be the standard reporting mechanism globally, dual reporting of a reasonable subset of laboratory tests in U.S. conventional units and SI units might be necessary to minimize conversion needs during review. Identification of units to be used for laboratory tests in clinical trials and solicitation of input from the review divisions should occur as early as possible in the development process. For more information, please see the FDA website entitled Study Data Standards Resources[7].

COMPOUNDED DRUG PRODUCT REQUIREMENTS

As described at 21 CFR 210.2(c), a drug product, including a compounded product, intended for use in a clinical study must be prepared in accordance with the current good manufacturing practice requirements appropriate for the product. For questions or clarification, contact [redacted].

STUDIES IN HUMANS MAY NOT BE CONDUCTED UNDER THIS PIND

Before you may conduct studies in humans, you must submit a full Investigational New Drug Application (IND, see 21 CFR Part 312) by amending this PIND with the required information. Include the above PIND number in Box 6 of the form FDA 1571 submitted with your IND. Send your IND submission in eCTD format through the ESG. Submissions that do not adhere to the requirements stated in the eCTD Guidance will be subject to rejection. For more information please visit FDA.gov.[8].

SECURE EMAIL COMMUNICATIONS

Secure email is required for all email communications from FDA when confidential information (e.g., trade secrets, manufacturing, or patient information) is included in the message. To receive email communications from FDA that include confidential information (e.g., information requests, labeling revisions, courtesy copies of letters), you must establish secure email. To establish secure email with FDA, send an email request to [redacted]. Please note that secure email may not be used for formal regulatory submissions to applications (except for 7-day safety reports for

[6] https://www.fda.gov/industry/study-data-standards-resources/study-data-submission-cder-and-cber
[7] http://www.fda.gov/ForIndustry/DataStandards/StudyDataStandards/default.htm
[8] http://www.fda.gov/ectd

INDs not in eCTD format).

PATIENT-FOCUSED ENDPOINTS

An important component of patient-focused drug development is describing the patient's perspective of treatment benefit in labeling based on data from patient-focused outcome measures [e.g., patient-reported outcome (PRO) measures]. Therefore, early in product development, we encourage sponsors to consider incorporating well-defined and reliable patient-focused outcome measures as key efficacy endpoints in clinical trials, when appropriate, and to discuss those measures with the Agency in advance of confirmatory trials. For additional information, refer to FDA's guidance for industry *Patient-Reported Outcome Measures: Use in Medical Product Development to Support Claims.*

PROSPECTIVE ASSESSMENTS OF SUICIDAL IDEATION AND BEHAVIOR IN CLINICAL PROTOCOLS

Treatment-emergent suicidal ideation and behavior have been identified as a concern for a number of drugs and drug classes. For example, meta-analyses of clinical trial data for both antiepileptic drugs and antidepressants have demonstrated that these drugs increase the risk of suicidal ideation and behavior. Spontaneous reports have led to similar concerns with other drugs as well, e.g., isotretinoin and other tretinoins, beta blockers, reserpine, smoking cessation drugs, and drugs for weight loss. Because of these concerns, a prospective assessment for suicidal ideation and behavior should be included, when appropriate and feasible, in clinical trials involving all drugs and biological products for neurological indications. These assessments should generally be included in every clinical protocol, at every visit, and in every phase of development, with the exception of single-dose trials in healthy volunteers. These assessments should be conducted whether or not a particular product is known or suspected to be associated with treatment-emergent suicidal ideation and behavior. A sponsor considering the omission of the assessment of suicidal ideation and behavior from a particular clinical protocol should prospectively discuss this omission with the Division of Neurology 1.

4.0 ISSUES REQUIRING FURTHER DISCUSSION

None.

5.0 ACTION ITEMS

None.

6.0 ATTACHMENTS AND HANDOUTS

The Sponsor's PowerPoint is appended to these minutes.

Development of a PrP-lowering divalent siRNA therapy for prion disease

Outline:

Background (5 minutes)

- Prion disease
- Therapeutic strategy
- PrP-lowering di-siRNA therapy

Questions and discussion topics (40 minutes)

2. **Toxicology plan**: Does the agency agree that data from these studies would be appropriate to support the proposed clinical plan?
3. **Drug product:** Does the agency agree that [our testing plan] is acceptable?

Summary and action items (10 minutes)

Goal of today's meeting

Our goal is to identify a path to submission of an **investigator-initiated IND** to administer a single dose of the **PrP-lowering divalent siRNA** molecule 2439-exNA to a **small number** of **symptomatic** prion disease patients.

We hope to solicit from FDA scientists feedback on:

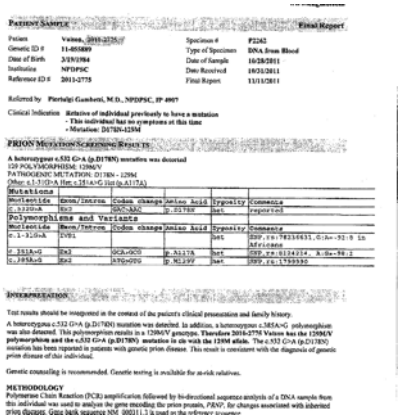
- Our proposed toxicology studies in rodents and non-rodents
- Our testing plan for drug product produced through sterile fill-finish

Background

0. Introduction to our team
1. Prion disease
2. Therapeutic hypothesis
3. PrP-lowering di-siRNA treatment extends survival of prion-infected mice

Introduction to our team

We – Eric Minikel and Sonia Vallabh – are a wife-husband team who became scientists after Sonia learned in 2011 that she had inherited a lethal genetic mutation in *PRNP* from her mother, who died of prion disease.



PATIENT SAMPLE — Final Report

Patient	Vallabh, [redacted]	Specimen #	P2262
Genetic ID #	11-005889	Type of Specimen	DNA from Blood
Date of Birth	3/19/1984	Date of Sample	10/28/2011
Institution	NPDPSC	Date Received	10/31/2011
Reference ID #	2011-2775	Final Report	11/11/2011

Referred by Pierluigi Gambetti, M.D., NPDPSC, IP 4907

Clinical Indication Relative of individual previously to have a mutation
- This individual has no symptoms at this time
- Mutation: D178N-129M

PRION MUTATION SCREENING REPORT

A heterozygous c.532 G>A (p.D178N) mutation was detected
129 POLYMORPHISM: 129M/V
PATHOGENIC MUTATION: D178N - 129M
(Mut: c.1-3:G>A Hm: c.381A>G Het (p.A117A))

Mutations

Nucleotide	Exon/Intron	Codon change	Amino Acid	Zygosity	Comments
c.532G>A	Ex2	GAC>AAC	p.D178N	het	reported

Polymorphisms and Variants

Nucleotide	Exon/Intron	Codon change	Amino Acid	Zygosity	Comments
c.1-3:G>A	IVS1			het	SNP rs79234631, G:Ac-52:0 in Africans
c.351A>G	Ex2	GCA>GCG	p.A117A	het	SNP rs:6124374, A:Ge-98:1
c.385A>G	Ex2	ATG>GTG	p.M129V	het	SNP rs:1799990

INTERPRETATION

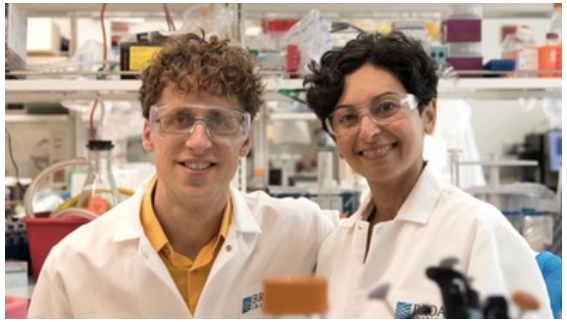
Test results should be interpreted in the context of the patient's clinical presentation and family history.

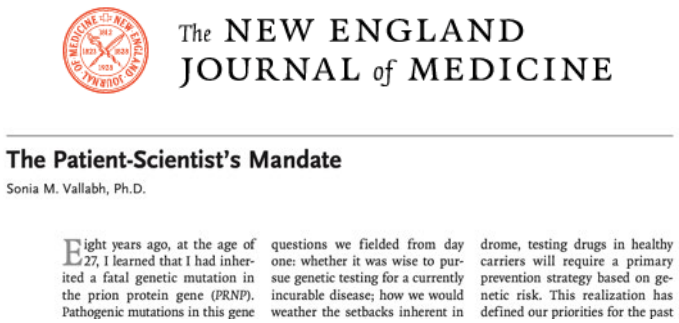
A heterozygous c.532 G>A (p.D178N) mutation was detected. In addition, a homozygous c.385A>G polymorphism was also detected. This polymorphism results in a 129M/V genotype. Therefore 2010-2775 Vallabh has the 129M/V polymorphism and the c.532 G>A (p.D178N) mutation in cis with the 129M allele. The c.532 G>A (p.D178N) mutation has been reported in patients with genetic prion disease. This result is consistent with the diagnosis of genetic prion disease of this individual.

Genetic counseling is recommended. Genetic testing is available for at-risk relatives.

METHODOLOGY

Polymerase Chain Reaction (PCR) amplification followed by bi-directional sequence analysis of a DNA sample from this individual was used to analyze the gene encoding the prion protein, *PRNP*, for changes associated with inherited prion diseases. Gene bank sequence NM_000311.3 is used as the reference sequence.





The NEW ENGLAND JOURNAL of MEDICINE

The Patient-Scientist's Mandate

Sonia M. Vallabh, Ph.D.

Eight years ago, at the age of 27, I learned that I had inherited a fatal genetic mutation in the prion protein gene (*PRNP*). Pathogenic mutations in this gene

questions we fielded from day one: whether it was wise to pursue genetic testing for a currently incurable disease; how we would weather the setbacks inherent in

drome, testing drugs in healthy carriers will require a primary prevention strategy based on genetic risk. This realization has defined our priorities for the past

Today, we co-run a lab at the Broad Institute devoted to developing a treatment in our lifetimes.

BROAD INSTITUTE

MASSACHUSETTS GENERAL HOSPITAL
NEUROLOGY

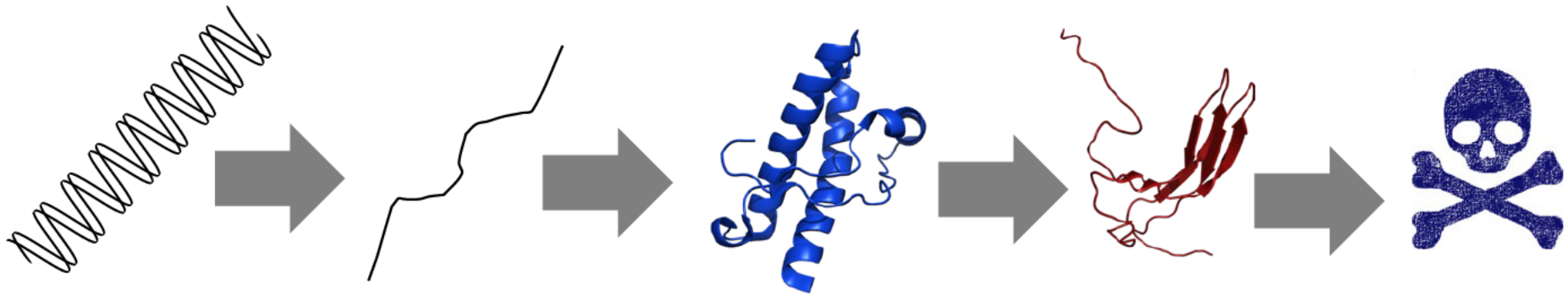
Reference ID: 5215048

Prion disease is a rapidly progressive dementia with no current standard of care

Median survival	5 months from first symptom 1-2 months from randomization in past trials
Prevalence	Causes 1 in 6,000 deaths
Etiology	85% sporadic (spontaneous; no known genetic or environmental cause) 15% genetic
Diagnosis	Premortem prion detection in CSF (>90% sensitive and specific)
Treatment	No standard of care ASO in preclinical development, but repeatedly delayed

Reference ID: 5215048

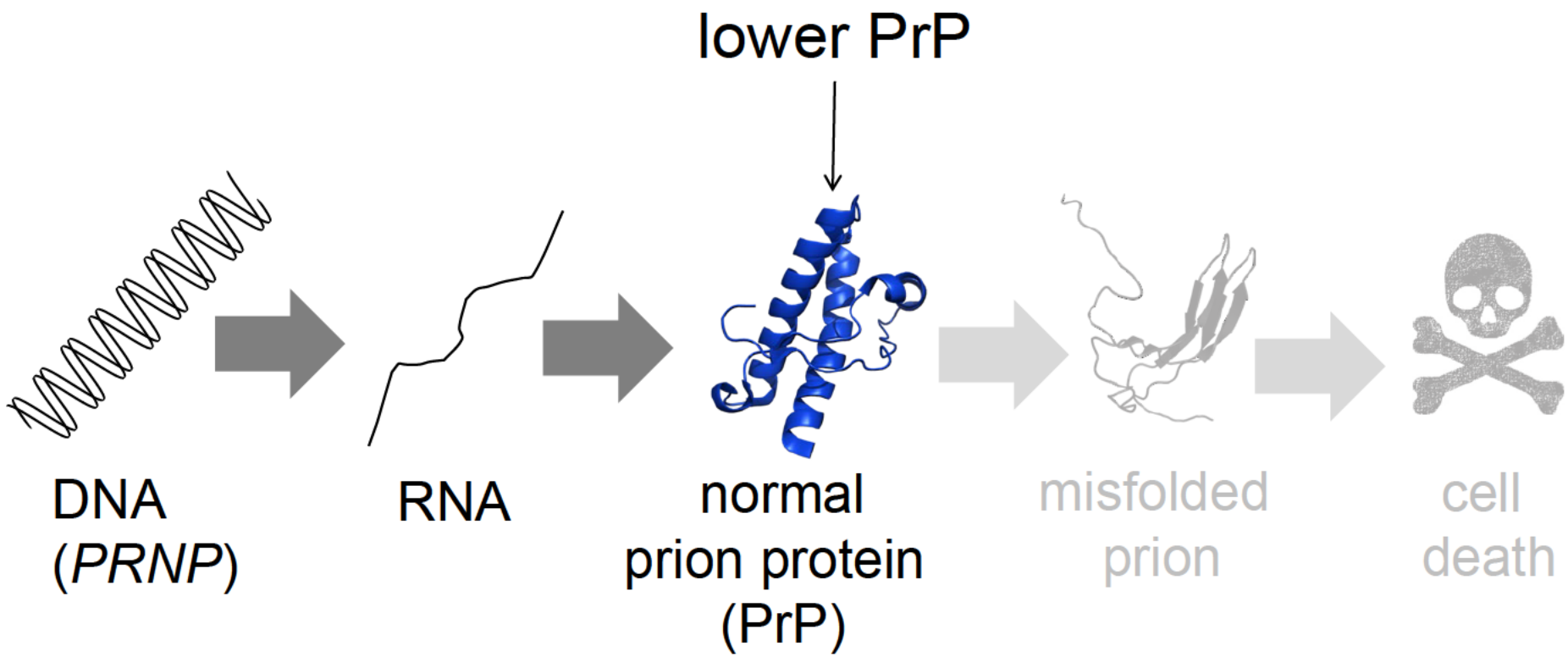
Our therapeutic strategy, lowering PrP, targets the sole root cause of all prion disease



DNA (*PRNP*) → RNA → normal prion protein (PrP) → misfolded prion → cell death

Reference ID: 5215048

Our therapeutic strategy, lowering PrP, targets the sole root cause of all prion disease



Reference ID: 5215048

The safety and efficacy of PrP lowering are well supported by genetic and pharmacologic data

Safety:

- PrP KO animals (multiple species) → HEALTHY
- Heterozygous loss-of-function humans → HEALTHY

Efficacy:

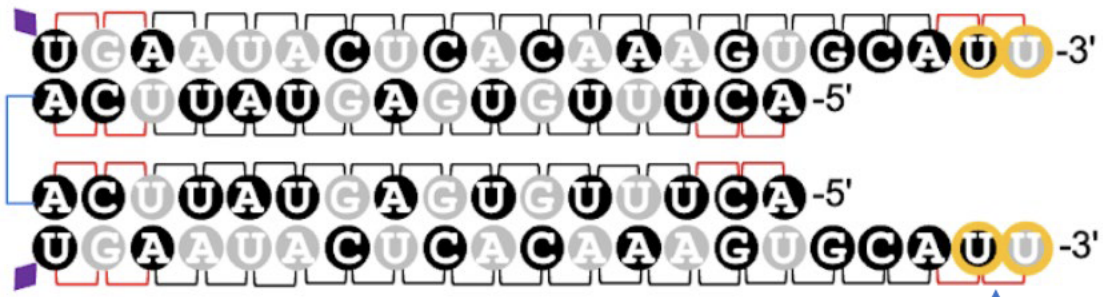
- PrP knockout (KO) animals (multiple species) → FULL PROTECTION
- Genetic or pharmacological PrP reduction → DOSE DEPENDENT PROTECTION
- Presymptomatic PrP reduction → EXTENDS HEALTHY LIFE
- Symptomatic treatment → SLOWS DISEASE

Reference ID: 5215048

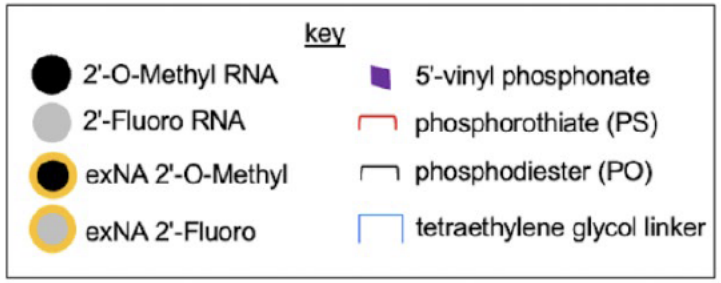
Divalent siRNA is a new modality with promising potency, durability and CNS distribution

Backbone = fully 2' modified

Divalent = 2 linked double stranded siRNAs

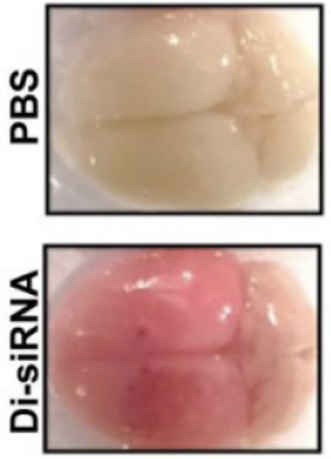


exNA modification



MW: ~27 kDa

- Potent target RNA knockdown
- Durability out to 6 months
- Broad CNS distribution
- Not yet tested in humans

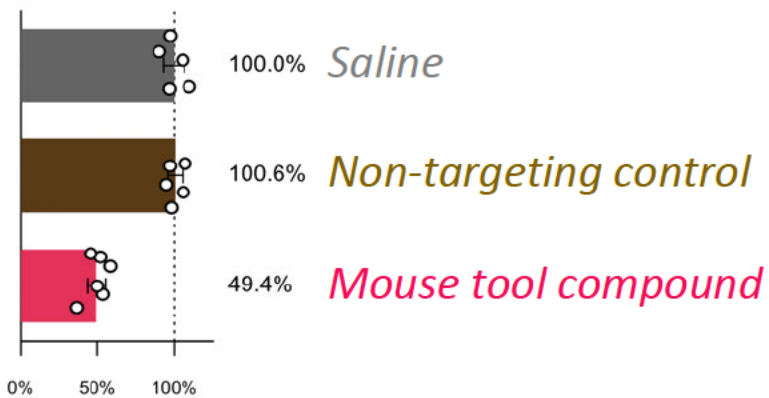


Alterman 2019, PMID: 31375812

Reference ID: 5215048

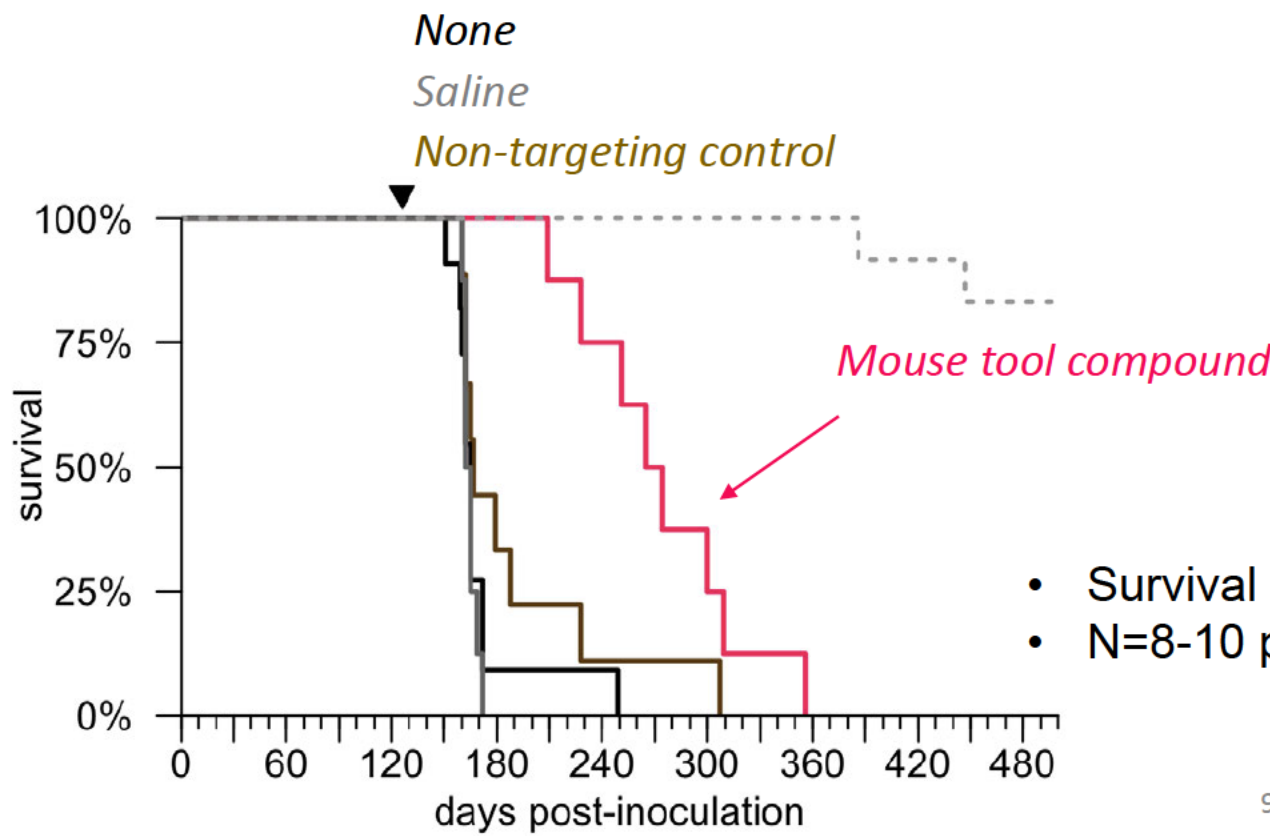
A single dose of PrP-lowering di-siRNA triples remaining survival time in prion-infected mice

Potent PrP knockdown, 1 month after a single dose (10 nmol, 270 µg)



- Wild-type mice
- Whole brain hemisphere PrP ELISA readout, N=6 per group

Survival study: symptomatic timepoint, single dose (10 nmol, 270 µg)

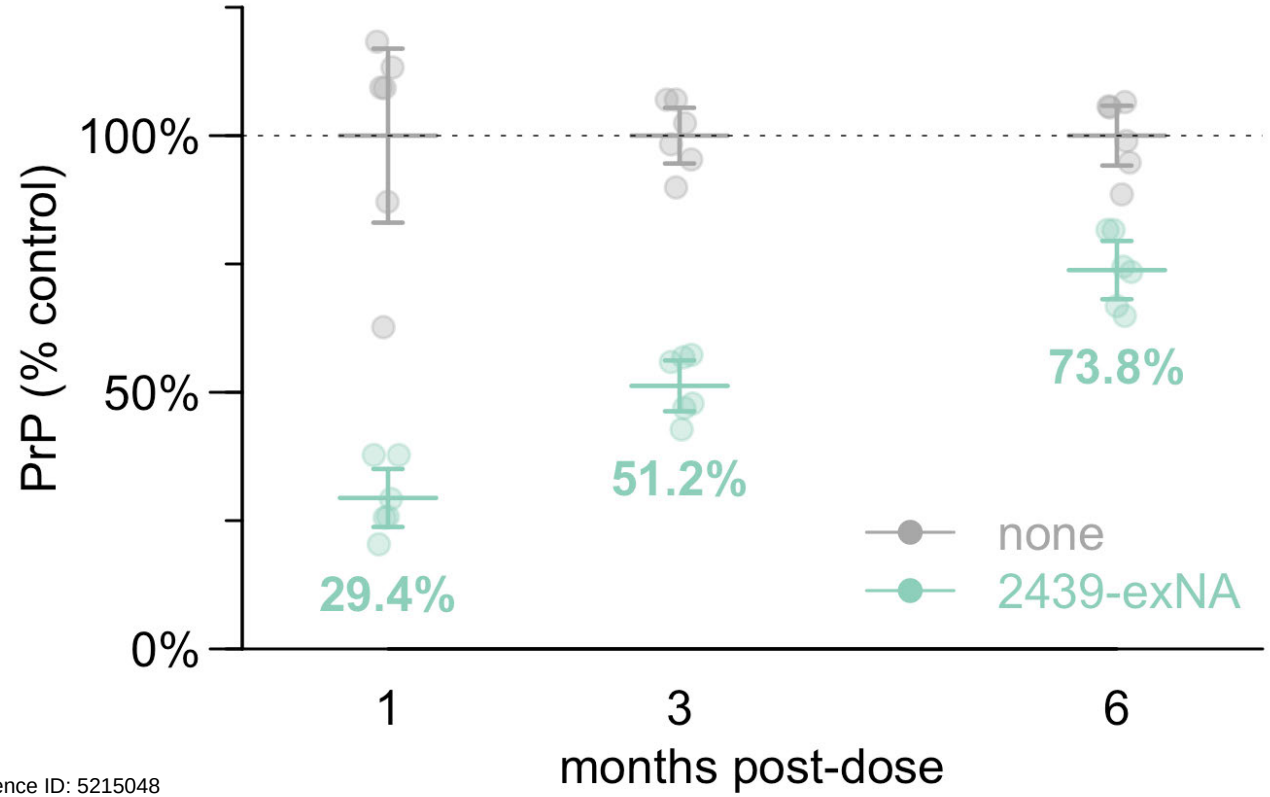


- Survival endpoint
- N=8-10 per group

Reference ID: 5215048

Our drug candidate is yet more potent, and is durable out to 6 months in humanized mice

- Dose: 5 nmol (~135 µg) ICV – half the dose on the previous slide
- Tg(huPrP) mice
- Whole brain hemisphere PrP ELISA readout, N=6 per group



Program structure:

- No commercial sponsor
- IND-enabling studies to be managed by Vallabh/Minikel lab at the Broad Institute
- Funding: NINDS URGenT U01 award (June 2023- May 2025)

Reference ID: 5215048

Questions and discussion topics

2. Toxicology plan
3. Testing of drug product

Reference ID: 5215048

We propose pivoting our plan to an Expanded Access IND

- Our briefing package was originally submitted in consideration of a standard IND
- An Expanded Access IND may be more appropriate
  - Uniformly fatal disease
  - No alternative treatments
  - No trials currently enrolling
  - Strong evidence in favor of PrP lowering as a therapeutic strategy, combined with reasonable nonclinical data to be gathered by the time of IND submission, would present a favorable risk/benefit balance to patients facing a certainty of dementia and death
- Primary goal would be treatment, patients would not be randomized
- Could begin as Intermediate Size or as Individual Patient and expand if/as appropriate

Reference ID: 5215048

[Nonclinical] We seek a nonclinical strategy that balances safety with an urgent unmet medical need

- On-target reduction of PrP is safe
  - Supported by genetic KO and LOF data from mice, rats, goats, cows and humans, collected over 40 years
- Off target drug effects will be modeled in rodents and non-rodents
- Any drug-related liver and kidney toxicity would be discriminable from prion disease
- We propose to dose only a small number of patients

Reference ID: 5215048

Question: Does the agency agree that data from [our proposed toxicology] studies would be appropriate to support the proposed clinical plan?

Our proposal:

We propose to conduct GLP toxicology and pharmacokinetic studies in rats, supplemented by non-GLP pharmacokinetic/pharmacodynamic studies in humanized mice and off-target analysis in human brain-derived cell lines.

FDA response:

Nonclinical safety testing is typically conducted in two species (rodent and nonrodent). In addition, we have concerns regarding the potential toxicity of divalent siRNA 2439- exNA, which has modifications (e.g., exNA) that may potentially increase its toxicity and for which you state there is no previous human experience. Therefore, to support the proposed clinical study of siRNA 2439-exNA, you will need to provide GLP-compliant toxicity studies of divalent siRNA 2439-exNA in **rodent and nonrodent.**

Reference ID: 5215048

We propose to conduct GLP nonclinical safety studies in rats

Clinical dosing route:
bolus delivery into CSF by intrathecal (IT) injection

Clinically relevant dosing regimen:
single dose

Outcomes:
- In-life monitoring
- Full necropsy for a standard battery of safety and toxicology endpoints (including CNS, cardiovascular, and respiratory) and brain tissue drug concentration determination

Group	N	Dose level	Harvest timepoint
1	10F/10M	0	1 month
2	10F/10M	300 µg	1 month
3	10F/10M	1 mg	1 month
4	10F/10M	3 mg	1 month

Note that dose levels shown are tentative, to be adjusted pending results of PD/PK modeling in humanized mice.

Reference ID: 5215048

We additionally propose to conduct GLP nonclinical safety studies in dog or sheep

Clinical dosing route:
bolus delivery into CSF by intrathecal (IT) injection

Clinically relevant dosing regimen: single dose

Outcomes:
- In-life monitoring
- Full necropsy for a standard battery of safety and toxicology endpoints (including CNS, cardiovascular, and respiratory) and brain tissue drug concentration determination.

Group	N	Dose level	Harvest timepoint
1	4M/4F	0	1 month
2	4M/4F	10 mg	1 month
3	4M/4F	30 mg	1 month
4	4M/4F	100 mg	1 month

Note that dose levels shown are tentative, to be adjusted pending results of PD/PK modeling in humanized mice.

Reference ID: 5215048

[CMC] We seek to maximize the ability of manufactured Drug Product to directly benefit patients, while ensuring adequate safety testing.

- 20g synthesis of di-siRNA human-targeting lead underway
- Goals for this batch of manufactured Drug Product:
  - Quality control testing
  - Proposed nonclinical studies
  - Proposed clinical studies
- Proposals to maximize amount of drug available for nonclinical and clinical:
  - Exemption from USP61/71/85 qualifications
  - Reduce fill volume for sampling vials
  - Reduce number of vials for certain tests

Reference ID: 5215048

Question: Does the agency agree that [our plan for drug product testing] is acceptable?

Our original proposal:

We proposed to generate Drug Product vials of different sizes, with smaller aliquots used for testing than dosing, to maximize utilization of manufactured drug.

FDA response:

- Your proposal may be acceptable for the initial study if the vials used for stability testing (and other quality control tests) can represent the **worst case in terms of the head space** - i.e., compared with the vials used for clinical dosing, such that the stability vials have higher headspace to drug product fill volume ratio. Final determination of the acceptability of such approach would be made at the time of the IND submission when all the supportive data may be evaluated.
- Please note that as the drug development continues, you should **use the same container closure system** for all the drug product vials.

Detailed proposal on following slides

Reference ID: 5215048

We propose to create vials with reduced fill volume for quality control testing

- 20 g DS
- Proposal: Formulate at 10 mg/mL in 100+ identical vials of 10 mL capacity, filled 2-5 mL
  - Clinical: 5.3 mL vials (USP1151 requires 5.3 mL fill to recover 5.0 mL)
  - Testing: 2 mL vials (USP71 has same requirements for any fill volume ≥2 mL in a 100-500 vial batch)

- Under the above plan, **testing vials will represent the worst-case scenario** in terms of head space (2 mL volume in 10 mL vial vs. 5 mL volume in a 10 mL tube for clinical use.)
- The same vials will be used for both clinical and testing units, with the **same container closure system.**

Reference ID: 5215048

We now aim to pivot away from compounding and toward sterile fill-and-finish

- We initially envisioned drug product being formulated through hospital compounding under USP797, which has been used for some N-of-1 programs
- We are hopeful that we have identified a safer option with a sterile fill-and-finish partner
- Following slides will propose a detailed plan

Reference ID: 5215048

We will partner with well-established sterile fill-finish CDMO Argonaut Manufacturing Services, which is experienced with N-of-few

D-U-N-S Site Identification Number: 080300725
FDA Facility Establishment Identifier (FEI Number): 3013508066

We will fit within Argonaut's established media fill program

- Has historically Included both commercial and N-of-few
- Our program would use the same machinery, operators and procedures as these other projects

Extensive prior data from other programs

- Company's position: can't share these data directly with us, but <u>can share with FDA on our behalf.</u>

Based on their state-of-the-art isolator/process and established track record, we propose that some flexibility in testing requirements for our program would be justified (details next slides).

Reference ID: 5215048

Specific reductions and exemptions would dramatically increase drug utilization in the clinic (slide 1/2)

all with 2 mL sampling vials

Topic	Proposal	Justification
1. Qualification/verification/validation (BNF) of USP71, 61, 85 methods for sterility, endotoxin, bioburden	Request exemption	As a class, oligonucleotides have not been known to cause growth inhibition or interfere with these tests. Argonaut uses isolator technology with vaporized hydrogen peroxide sterilization of manufacturing equipment, virtually eliminating opportunities for microbial contamination. We will fit within their established processes and media fill program.
2. Sterility	Reduce to 1 vial	Our patient population and number of patients is consistent with an "N-of-few" risk profile while employing state-of-the-art isolator fill-and-finish that is significantly lower risk than USP797 hospital compounding pharmacies.
3. Particulates	Reduce to 1 vial	The manufacturing process, using all-plastic flow path and minimal metal, confers little risk of particulates. In the unlikely event of problems, as the batch is homogeneous and small, 1 vial will be representative.
4. In-process bioburden	Request exemption	Manufacturing processes for oligonucleotide products use organic rather than aqueous chemistry and are not conducive to microbial contamination. Additionally, DS will already be tested per USP61 for bioburden on release, minimizing the probability of bioburden sufficient to compromise the redundant sterilizing filters.
5. Stability	Request exemption from in-use stability study; request to do limited stability study at 6, 12, 24 months at -20°C	Oligonucleotides as a class are highly stable, and proposed nonclinical and clinical studies have a compact timeline.

Reference ID: 5215048

Specific reductions and exemptions would dramatically increase drug utilization in the clinic (slide 2/2)

all with 2 mL sampling vials

Topic	Proposal	Justification
6. Identity	Perform per standard IND	
7. Concentration/Assay	Perform per standard IND	
8. pH	Perform per standard IND	
9. Osmolality	Perform per standard IND	
10. Purity	Perform per standard IND	
11. Endotoxin	Perform per standard IND	

Reference ID: 5215048

Discussion, summary and action items

Thank you for your time

This is a representation of an electronic record that was signed electronically. Following this are manifestations of any and all electronic signatures for this electronic record.

/s/

LAURA A JAWIDZIK
07/27/2023 05:14:09 PM
Signing on behalf of Emily Freilich